



A Comparison of Percutaneous Pedicle Screw Accuracy Between Robotic Navigation and Novel Fluoroscopy-Based Instrument Tracking for Patients Undergoing Instrumented Thoracolumbar Surgery

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■ BACKGROUND: The accuracy of pedicle screws placed with instrument tracking and robotic navigation are individually comparable or superior to placement using standard fluoroscopy, however head-to-head comparisons between these adjuncts in a similar surgical population have yet to be performed.

■ METHODS: Consecutive patients undergoing percutaneous thoracic and lumbosacral spinal instrumentation were retrospectively enrolled. Instrumentation was performed using either fluoroscopy-based instrument tracking system (TrackX, TrackX Technologies) or robotic-navigation (ExcelsiusGPS, Globus Medical). Postinstrumentation computed tomography scans were graded for breach according to the Gertzbein-Robbins scale, with “acceptable” screws deemed as Grade A or B and “unacceptable” screws deemed as Grades C through E. Accuracy data was compared between both instrumentation modalities.

■ RESULTS: Fifty-three patients, comprising a total of 250 screws (167 robot, 83 instrument tracking) were included. The overall accuracy between both modalities was similar, with 96.4% and 97.6% of screws with acceptable accuracy between instrument tracking and robotic navigation, respectively (I-squared 0.30, df = 1, $P = 0.58$). Between instrument tracking and robotic navigation, 92.8% and 95.8% of screws received Grade A, 3.6% and 1.8% a Grade B, 1.2% and 1.2% a Grade C, 1.2% and 0.6% a Grade D, and 1.2% and 0.6% a Grade E, respectively. The robot was abandoned intraoperatively in 2 cases due to unrecoverable registration

inaccuracy or software failure, leading to abandonment of 8 potential screws (4.8%).

■ CONCLUSIONS: In a similar patient population, there is a similarly high degree of instrumentation accuracy between fluoroscopy-based instrument tracking and robotic navigation. There is a rare chance for screw breach with either surgical adjunct.

INTRODUCTION

Pedicle screw fixation remains the biomechanical standard for thoracic and lumbosacral instrumented fusion due to its ability to span all 3 columns of the spine.¹ Properly placed pedicle screws allow for rigid fixation which facilitates fusion and internal bracing. Improperly placed pedicle screws, however, result in biomechanically inferior constructs while also risking injury to critical neural or neurovascular structures. In these cases, significant and sometimes irreversible neural injury can occur, and screw loosening in biomechanically inferior constructs can result in pseudarthrosis. These complications oftentimes require revision surgery which subsequently decreases quality of life and can incur legal repercussions.² As a result, accurate placement of thoracic and lumbosacral pedicle screws is of paramount importance to instrumented spine surgery.

Pedicle screws placed with conventional fluoroscopy, alone, have highly variable accuracy rates. In a systematic review by Mason et al., only 2532 of 3719 (68.1%) were found to be without breach.³ Contrarily, other reports show fluoroscopy-based accu-

Key words

- Accuracy
- Fluoroscopy
- Instrument navigation
- Percutaneous
- Robot

Abbreviations and Acronyms

CT: Computed tomography

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racy rates as high as 94%.⁴ The ultimate desire to completely eliminate pedicle screw misplacement led to the development of many assistive technologies built to increase pedicle screw accuracy. This ushered in a new era of navigated spinal instrumentation currently led by robotic and 3-dimensional intraoperative computed tomography guidance.⁵ While there was a proven learning curve associated with adoption of these technologies, surgeons soon recognized their value at improving pedicle screw accuracy compared to traditional free-hand and fluoroscopy techniques.⁶⁻⁸ As with all new technologies, however, these assistive devices were associated with significant costs and were not entirely risk-free. While the competitive market has driven down overall costs, many forms of robotic and intraoperative computed tomography (CT) navigation require significant initial investments. Depending on the system, this investment could potentially be upward of \$2 million.⁹ Additionally, this underestimates the cost of owning and utilizing each technology, as this figure does not account for instrumentation consignments and other disposable fees associated with each case. These barriers have prevented widespread adoption of these devices, especially at smaller-volume hospitals or surgical centers.

Alternative fluoroscopy-based instrument tracking (TrackX, TrackX Technologies, LLC., NC, USA) has been previously demonstrated as a safe and efficient surgical adjunct for a variety of surgical tasks including pedicle screw instrumentation. In contrast to robotic or CT-guided navigation, TrackX instrument-tracking augments already-existing standard c-arm fluoroscopy to provide surgeons with pseudo-live biplanar fluoroscopy using a single c-arm. Compared to standard fluoroscopy, this technology has been shown in clinical trials to reduce radiation exposure, number of x-rays required to perform fluoroscopy-intensive tasks, and operative time.^{10,11} While complication rates associated with this technology are low, the accuracy at which it can perform surgical instrumentation, particularly pedicle screw instrumentation, is currently unknown. Further, no head-to-head comparison has ever compared the accuracy of pedicle screws between fluoroscopy-based instrument tracking and robotic navigation amongst the same surgical population. As such, the authors sought to compare thoracic and lumbosacral pedicle screw accuracy between robotic navigation and fluoroscopy-based instrument tracking in patients undergoing minimally invasive thoracic and lumbosacral fixation.

MATERIALS AND METHODS

Study Design

The authors performed a retrospective review of patients meeting the following inclusion criteria: 1) age 18 years or greater, 2) undergoing percutaneous thoracic or lumbosacral pedicle screw insertion using either robotic instrumentation with the ExcelsiusGPS system using Globus Creo minimally invasive screws (Globus Medical, Audubon, PA, USA) or instrument-tracking with TrackX (TrackX Technologies, LLC, NC, USA) using ViperPrime single-step percutaneous screws (DePuy Synthes, Warsaw, IN, USA), and 3) postinstrumentation CT lumbar spine without contrast. All patients underwent minimally invasive transforaminal lumbar interbody fusion, percutaneous lumbar interbody fusion,

lumbar lateral interbody fusion, prone transposas interbody fusion, or lone percutaneous pedicle instrumentation for internal bracing. This was a single surgeon study, and the surgeon who performed the above procedures was fellowship trained in spine surgery with 5 years of experience following fellowship. Surgical indications included degenerative or isthmic spondylolisthesis or traumatic thoracolumbar unstable fractures requiring internal bracing. Patients were excluded if they underwent conversion to open pedicle screw instrumentation or if they did not have post-instrumentation CT imaging. The decision to use robotic or instrument-tracking to facilitate screw placement was based on surgeon's preference as well as availability of technology. Accuracy was evaluated using intraoperative or postoperative CT scans and the reviewer was blinded to the imaging modality. Screw accuracy was graded using the widely validated Gertzbein-Robbins accuracy scale. An additional review of all screws, blinded to the assistive technology, was performed by the same reviewer 6 weeks following initial review. This established an intrarater reliability score.

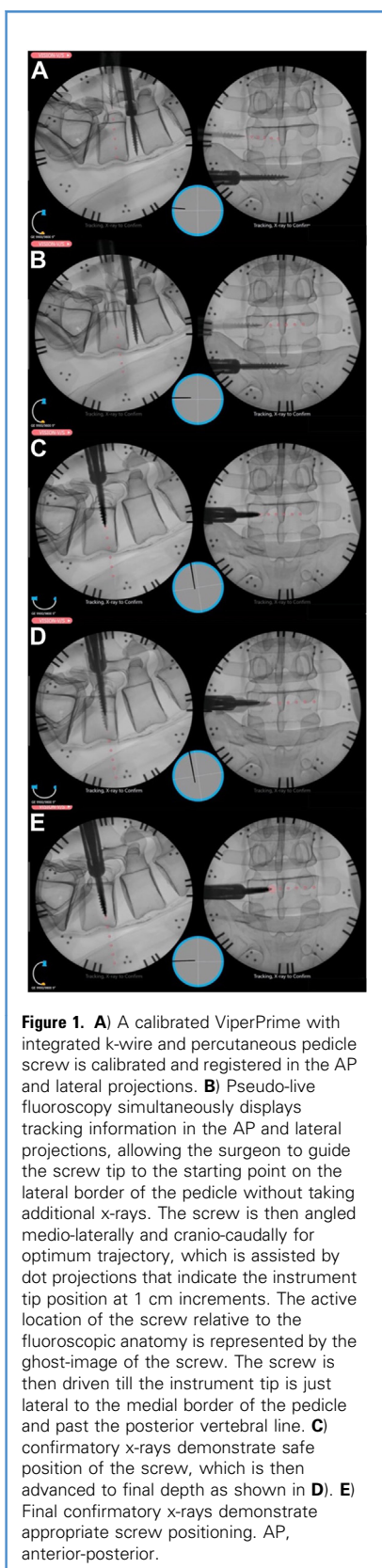
Screw grades were assigned based on pedicle wall breach, with Grade A demonstrating no breach, Grade B demonstrating 0.1–1.9 mm breach, Grade C demonstrating 2.0–3.9 mm breach, Grade D demonstrating 4.0–5.9 mm breach, and Grade E with 6.0 mm breach or greater. Screws with Grade A or B designation were categorized into the “acceptable screw” cohort while Grade C, D, or E screws were assigned to the “unacceptable screw” cohort. Intraoperative and postoperative complications including screw removal, omission or screw or aborted technology, or revision surgery were recorded. Institutional review board approval was obtained. Patient consent was not needed as this study was retrospective in nature with no risk of harm to the patient.

Statistical Analysis

Screw accuracy by individual Grade and acceptability cohort were then compared between robotic navigation and instrument tracking using student's t-test while categorical data were compared with Fisher's exact t-test. χ^2 without Yates correction was used to compare final accuracy data with α 0.05. Study power was met based on an estimated mean of $97 \pm 2\%$ and $95 \pm 2\%$ between robot-guided and instrument-tracked screw accuracy, which were determined based on literature reported rates of screw accuracy and preliminary patient data, respectively. Enrollment ratio was initially set at 1:1; however based on actual enrollment, a ratio of 2:1 was used.

Robotic-Navigation Pedicle Screw Technique

To facilitate robotic pedicle screw instrumentation, a pre-instrumentation high-resolution CT of the instrumented levels was required. This was performed either 1 of 2 ways. Patients either underwent an elective preoperative CT lumbar spine without contrast with 1 mm slice thickness or underwent pre-instrumentation intraoperative CT lumbar spine with the mobile CT scanner (AIRO, BrainLab, Munich, Germany). Intraoperative anterior-posterior and lateral fluoroscopic images were then taken and fused to the CT. Screw trajectories were then planned, and the robot was draped and introduced to the sterile field. Screws were placed in sequential order, starting with 1) skin stab incision and followed by, 2) burr drill for pilot hole creation, 3) undersize screw



tap, and lastly 4) placement of pedicle screw. Intermittent intra-operative fluoroscopy prior to steps 2 and 3 confirmed robot accuracy prior to proceeding to the next step.

Instrument-Tracking Pedicle Screw Technique

For screws placed with instrument tracking, the c-arm was outfitted with a tracking collar over the image intensifier and a tracking cap over the radiation source. An infrared optical camera was placed at the foot of the patient, roughly 6 feet high and 6 feet from the working area. The tracking screen took the position of the standard fluoroscopy base and was typically directly across from the primary surgeon. After securing an optical tracking snap the ViperPrime screwdriver, the instrument is calibrated to screw length and registered in both the anterior-posterior and lateral planes. Subsequent instrument movements are shown as simultaneous biplanar pseudo-live fluoroscopy with dot projections indicating the projected instrument tip in 1 cm increments, which helps guide placement of instrument at the correct starting point and with the correct angulation and trajectory. The surgeon then uses this assistive information to place pedicle screws (Figure 1).

RESULTS

A total of 53 patients (29 robotic navigation, 24 instrument tracking) comprising 250 screws (167 robotic navigation, 83 instrument tracking) met inclusion criteria analysis. Study patients were accrued from a period from February 2019 through February 2021, which is the time period that the surgical robot and instrument tracking were available for use at the authors' institution. Average age, gender distribution, and body mass index did not differ between instrumentation methods and is summarized in Table 1.

In general, placement accuracy was high and consistent amongst both technologies. Using instrument tracking 92.8% of screws achieved Gertzbein-Robbins Grade A, while 3.6%, 1.2%, 1.2% (medial), and 1.2% received Grades B, C, D, and E, respectively. Using robotic navigation, 97.6% screws achieved Gertzbein-Robbins Grade A, while 1.8%, 1.2%, 0.6%, and 0.6% screws achieved Grades B, C, D, and E, respectively. Screw distribution amongst individual vertebral levels is summarized in Table 2 and accuracy grading is summarized in Table 3.

When further separated into *acceptable screw* and *unacceptable screw* designations based on Grade A + B versus Grade C + D + E, respectively, there was no statistical difference in accuracy between instrumentation modality ($I^2: 0.30$, $df = 1$, $P = 0.58$, Table 4). For the robot, there was a total of 4 unacceptable screws, with one at each of the following vertebral levels: T7, T8, T11, and S1. When using instrument tracking, there was a total of 3 unacceptable screws with 1 occurring at L3 and the remaining 2 occurring at L4. For both modalities, there was a high degree of accuracy. Following secondary review of data, intrarater reliability was established at 0.96, with no deviations in Gertzbein-Robbins Grade of more than 1 category and no cross-over between *acceptable* and *unacceptable* cohorts.

Complications

There was 1 Grade C and 1 Grade E screw for patients undergoing instrumentation using fluoroscopic instrument tracking. These

Table 1. Demographic Information

	Total	Instrument Tracking	Robot	P
Number of patients	53	24 (45.3%)	29 (54.7%)	-
Number of screws	250	83 (33.2%)	167 (66.8%)	-
Age (years)	61.3 ± 11.3	57.9 ± 11.9	62.0 ± 10.4	0.592
Male	27 (50.9%)	15 (62.5%)	12 (41.4%)	0.125
BMI	30.1 ± 5.3	30.4 ± 5.6	29.8 ± 5.1	0.733
BMI, Body mass index.				

patients underwent percutaneous L3-5 lumbar fusion and L3-S1 percutaneous lumbar fusion with final intraoperative fluoroscopic films demonstrating no egregious medialization of any screws. The angle of the left L4 screw in each case suggested possible medial breach, however electromyography monitoring did not demonstrate nerve irritation during placement, and stimulation of the screw after placement did not illicit any nerve root activation up to 20 mA. Postoperatively, however, the patients awoke with left knee extension weakness and radiculopathy. A postoperative CT of one of the patient's lumbar spine demonstrated the medialized screws, and they returned to the operating room that evening for screw removal (Figure 2). Both patients awoke at their previous baseline neurological exam and afterward experienced an uneventful recovery. The single Grade D screw was laterally breached on postoperative CT but the patient was asymptomatic and thus the screw was not revised. Of note, intraoperative postinstrumentation CT was utilized for every case, thereafter. There were no instances where instrument tracking needed to be abandoned. There were some cases where instrument tracking was used for 1 side of the construct while standard

Table 3. Gertzbein-Robbins Grading for Screws Placed Using Instrument Tracking Versus Robotic Navigation

	Instrument Tracking	%	Robot	%
Gertzbein-Robbins A	77	92.77%	160	95.81%
Gertzbein-Robbins B	3	3.61%	3	1.80%
Gertzbein-Robbins C	1	1.20%	2	1.20%
Gertzbein-Robbins D	1	1.20%	1	0.60%
Gertzbein-Robbins E	1	1.20%	1	0.60%
Total	83	100.00%	167	100.00%

fluoroscopy was used for another (resident and attending performing simultaneous instrumentation).

The 4 cumulative Grade C, D, and E screws placed with robotic instrumentation were all noticed on intraoperative post-instrumentation CTs and were spread across 2 cases. In both cases, the 2 screws were breached laterally and 2 were breached medially and all were part of a longer construct. Prior to leaving the operating room, they were removed without replacement (Figure 3). The robot was abandoned in 2 (6.9%) cases due to robot inaccuracy in 1 case and software failure in another case. In these 2 cases, instrumentation was completed with standard fluoroscopy. This amounted to a total of 8 (4.8%) screws.

DISCUSSION

The accurate placement of thoracolumbar and sacral pedicle screws has clear clinical benefit. Misplaced pedicle screws are not only biomechanically inferior but also may impinge on critical neural structures such as nerve roots or the spinal cord. Misplaced screws can cause neurological or painful symptoms and are a cause for revision surgery. For every revision surgery, there is an associated cost between \$4000–\$140,000 depending on need for revision surgery or disability resulting from nerve damage.¹² Misplaced screws resulting in symptomatic neurological changes or revision surgery are also frequently the subject of legal action, with the average spine surgery payout of \$1.2 million.² Accordingly, there has been a recent push to advance technologies which improve pedicle screw accuracy.

In this study, the authors demonstrate high accuracy of pedicle screw instrumentation with robotic navigation and fluoroscopy-based instrument-tracking. While both modalities had >95% acceptable screw accuracy grading, there were rare but persistent cases of significant breach using both technologies. As such, it remains important to validate instrumentation with a combination of intraoperative fluoroscopy, screw stimulation, and ideally postinstrumentation, intraoperative CT imaging.

To date, there is consensus that pedicle screws placed with robotic navigation are more accurate than free-hand pedicle screw instrumentation and those placed with standard fluoroscopy.¹³ In a review of 169 patients undergoing a total of 880 screws, Molliqaj et al. found a statistically higher rate of acceptable screw accuracy when using robotic navigation compared to free-hand placement

Table 2. Screw Placement per Instrumented Level

	Instrument Tracking	Robot
T7	0	2
T8	0	2
T9	0	4
T10	2	2
T11	3	2
T12	3	4
L1	3	4
L2	8	11
L3	13	15
L4	25	40
L5	23	49
S1	3	32
Total	83	167

Table 4. Comparison of Screw Acceptability Between Instrument Tracking and Robotic Navigation

	Instrument Tracking		Robot	
		%		%
Acceptable (A or B)	80	96.39%	163	97.60%
Unacceptable (C, D, E)	3	3.61%	4	2.40%

(93.4% vs. 88.9%, $P = 0.005$).¹⁴ Using the ExcelsiusGPS robot, a geometric accuracy study performed by Godzik et al. demonstrated a 3.4% pedicle screw breach rate with mean 2D accuracy of 2.0 mm with average angular offset of 5.6° compared to preoperative screw plans.¹⁵ In a study of 106 patients and 630 screws placed by the ExcelsiusGPS, Wallace et al. found a 1.5% rate of screw revision with 98.2% of screws rated either Gertzbein-Robbins Grade A or B.¹⁶ A 2020 meta-analysis by Li et al. demonstrated statistically similar screw accuracy between pedicle screws placed using the Mazor Renaissance robot and standard fluoroscopy,¹⁷ which was corroborated by a separate meta-analysis by Peng et al.¹⁸

Compared to robotic navigation, instrument tracking is more novel, and thus the impact of this assistive technology on pedicle screw accuracy is hereto unknown. This manuscript demonstrated a 96.4% rate of acceptable screw accuracy, which statistically rivals that of screws placed using a multitude of robotic navigation modalities including the ExcelsiusGPS system as was used in this

manuscript¹⁹ and is increased compared to screw accuracy from the surgeon's own standard fluoroscopy control group (88.9%). This comparison is consistent with the literature-reported accuracy rate of thoracolumbar pedicle screw instrumentation performed using unassisted fluoroscopy.²⁰

Furthermore, of the 29 patients who underwent pedicle screw instrumentation using robotic navigation, 2 (6.9%) patients required intraoperative abandonment of the robot. The inability to actively reregister the robot to the patient's anatomy, inadvertent manipulation of reference array markers, and inability to easily confirm instrumentation accuracy are additional sources of error that have contributed to the ~10% rate of intraoperative abandonment of robotic navigation as reported in the literature and as seen in the 2 abandoned cases from this study.^{15,21}

Intraoperative abandonment of robotic navigation results in significant financial waste and decreased operative efficiency. As medical expenditure becomes increasingly important in value-based care, hospitals are pushing to optimize patient care while lowering overhead costs. Robotic navigation has traditionally required high initial investment and additional costs associated with disposables and instrument contracts, which has been one of the major criticisms of robotic technology. Mazor's Renaissance, for instance, can cost \$550,000, plus an additional \$1500 in disposables per case and \$55,000 in annual maintenance and service.²² The ROSA spine robot is estimated to cost \$700,000, and the estimated initial investment for ExcelsiusGPS totals approximately \$1,000,000–\$1,500,000.⁵ While these prices are somewhat negotiable among hospital systems, they remain prohibitively expensive for many surgery

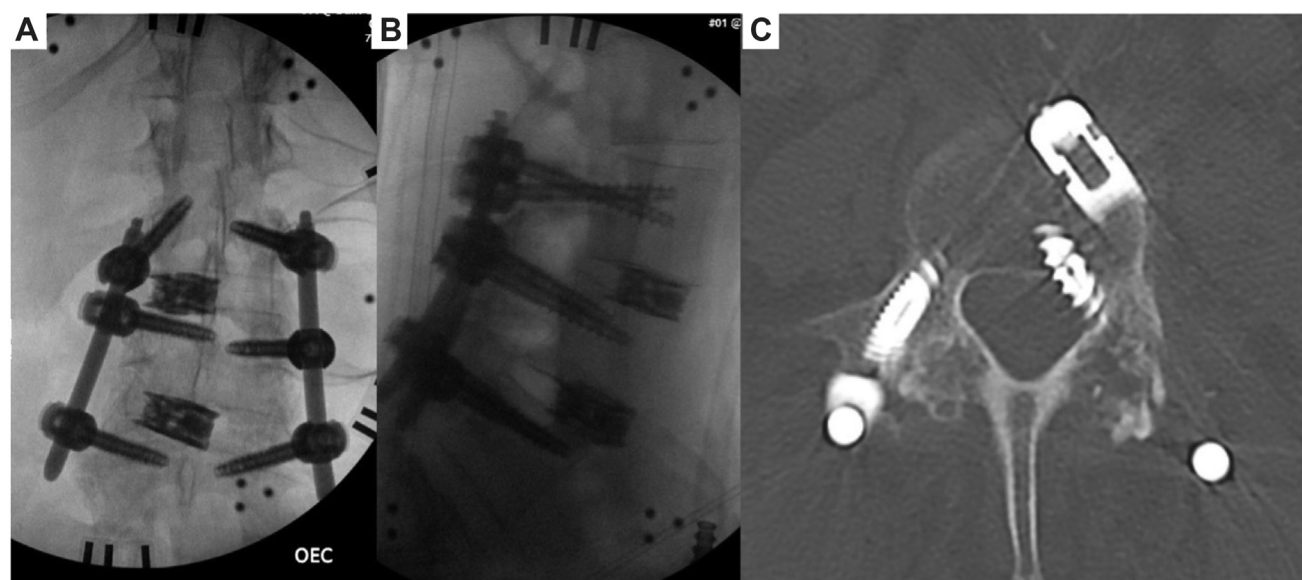
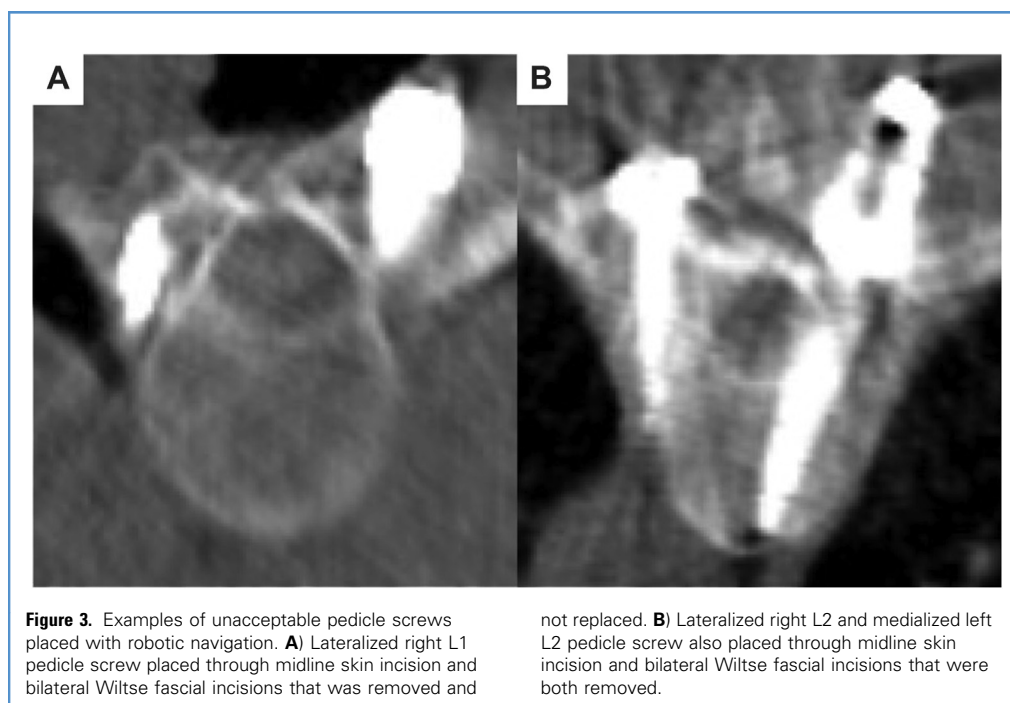


Figure 2. A) anterior-posterior and B) lateral intraoperative fluoroscopic films following L3-5 percutaneous lumbar interbody fusion. C) postoperative CT demonstrating medialization of the left L4 screw. On the anterior-posterior films, it is noted that the last several millimeters of the left L4 screw are

crossing midline in the AP projection while less than halfway through the vertebral body in the lateral view, which would suggest that the screw is medialized. CT, computed tomography; AP, anterior-posterior.



centers. Instrument tracking (TrackX) that utilizes the standard c-arm fluoroscope is priced on a per-case basis with no initial investment cost or instrumentation consignment, which contrasts the high investment cost needed for robotic navigation systems. To date, there are few studies on overall cost-effectiveness of both technologies, but available literature suggests that robotic navigation results in high total costs and higher costs per quality-adjusted life year.⁹

There are several limitations to this study. As patients were not randomized to instrumentation with either robotic navigation or instrument tracking, there could be differences related to patient factors such as body habitus and bone quality, however, demographic factors such as patient age and body mass index were statistically similar between groups. Additionally, as a single-surgeon study, there may be differences in learning curve that may differ amongst the surgical community. Lastly, this study did not quantify radiation exposure or the time it took to place screws individually. However, both modalities have been shown in previous studies to result in decreased radiation exposure to at least the surgeon and thus were not a primary endpoint for this study.^{10,11,23,24} Similarly, our groups prior work has revealed that specifically for minimally invasive transforaminal lumbar interbody fusions performed with standard fluoroscopy, those done with robotic-assistance consistently increased operative times while instrument-tracking was associated with both a short learning curve and improved operative times.^{11,25,26} Looking

forward, there is a need for additional studies comparing the efficiency of robot versus instrument tracking to see if there is a true “cutoff” where one technique is more efficient than the other, specifically for operative time. Nonetheless, this is the first manuscript suggesting high and equivalent pedicle screw accuracy between instrument tracking and robotic navigation.

CONCLUSION

In a similar patient population, there is a similarly high degree of instrumentation accuracy between fluoroscopy-based instrument tracking and robotic navigation. While meant to assist surgeons with screw placement, there is a rare but persistent chance for screw breach with either surgical adjunct. Both surgical adjuncts can effectively increase pedicle screw accuracy relative to literature-quoted, free-hand, and unassisted-fluoroscopy accuracy rates.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Timothy Wang, Eric Sankey, Vikram Mehta and Muhammad Abd-El-Barr. The first draft of the manuscript was written by Timothy Wang and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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